

Causal Inference in Economics and Marketing

Section 06: Regression Discontinuity and Kink Designs

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1 A rule creates an experiment

1.1 Thresholds are everywhere

A scholarship goes to students whose test score is at least 80. A party holds a seat when its vote share passes 50 percent. A class is split in two when enrollment exceeds 40. Unemployment benefits replace 55 percent of past earnings up to a cap. A house gets an energy label B rather than C when its score reaches 69. In each case a rule maps a continuous variable, the running variable, into a treatment through a cutoff.

A regression discontinuity (RD) design uses that rule. Units just below the cutoff and units just above it are alike in everything that varies smoothly with the running variable; they differ only in the treatment. Comparing them is an experiment the rule ran for us.

1.2 The estimand and why it is local

Let X be the running variable, c the cutoff, and $D = 1\{X \geq c\}$ the treatment in a sharp design. The target is

$$\tau_{RD} = E[Y(1) - Y(0) \mid X = c],$$

the average effect for units at the cutoff. Nothing in the design speaks about units far from c . That is the trade the design makes: the strongest internal validity among quasi-experimental designs (Dehejia's phrase) for an estimand that lives at one point. Section 8 shows what extra assumptions buy effects away from the cutoff.

1.3 One design, six figures

Every RD paper shows the same figures. The outcome jumps at the cutoff; predetermined covariates do not; the density of the running variable does not. We simulate the design of Lee (2008) as calibrated by Calonico, Cattaneo, and Titiunik (2014): a smooth outcome function with a jump of 0.04 at zero.

```
dat <- sim_rd(n = 3000, dgp = "covariates", seed = 1)
attr(dat, "tau_true")
#> [1] 0.04
```

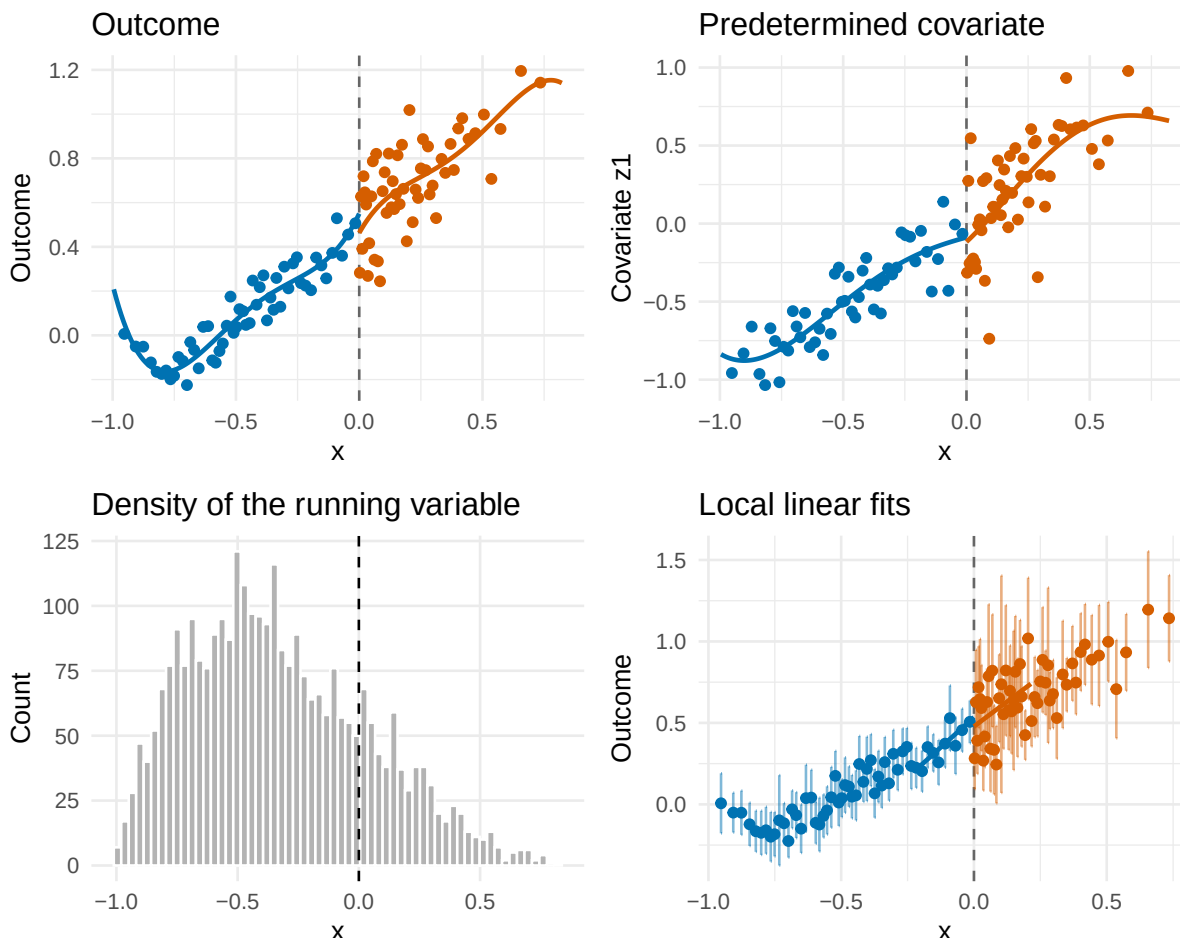


Figure 1: The three RD figures on a simulated design. Top left: the outcome jumps at the cutoff (binned means with fourth-order polynomial fits per side). Top right: a predetermined covariate does not jump. Bottom left: the density of the running variable is continuous. Bottom right: the same outcome plot with local linear fits inside the MSE-optimal bandwidth, the estimator’s own view of the data.

`rd_plot()` draws these as ggplot objects: bins are the IMSE-optimal quantile-spaced bins of Calonico, Cattaneo, and Titiunik (2015), the fits are polynomials per side or local linear fits inside a bandwidth, and any layer can be added with `+`.

1.4 Where naive approaches fail

Three ways to estimate the jump suggest themselves: compare means in a narrow window, fit one polynomial to the whole range on each side, or fit a line to the observations near the cutoff. The second was the original practice and is now discouraged (Gelman and Imbens 2019). The experiment below fits five estimators to 200 samples of size 1,000 from the Lee design.

Table 1: Five estimators of the jump on the Lee design, true effect 0.04, $n = 1,000$, 200 simulations. Local estimators use the MSE-optimal bandwidth of `rdbwselect()` and conventional standard errors; global polynomials use HC1.

Estimator	Bias	SD	RMSE	Mean SE	95% coverage
Global polynomial, order 4	0.0281	0.0427	0.0511	0.0410	0.85
Global polynomial, order 2	-0.0166	0.0271	0.0318	0.0262	0.89
Local linear, MSE bandwidth	0.0189	0.0414	0.0455	0.0365	0.87
Local linear, half bandwidth	0.0053	0.0547	0.0548	0.0504	0.91
Local quadratic	0.0019	0.0573	0.0572	0.0527	0.94

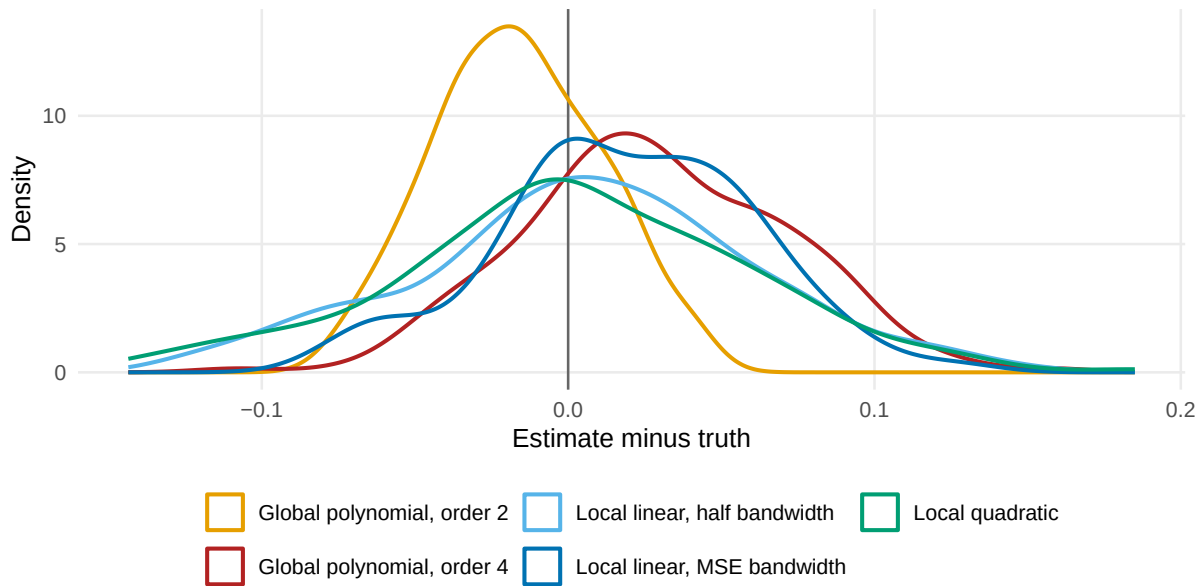


Figure 2: Sampling distributions of the five estimators, centered at the truth.

Three things are visible.

- The global polynomials report standard errors that are 0.96 (quartic) and 0.96 (quadratic) times their actual sampling standard deviations, with coverage 0.85 and 0.89. Their bias depends on whether the chosen order happens to match the curvature of the truth, which is unknown; Gelman and Imbens (2019) show that high-order fits also give observations far from the cutoff large weights on the estimate at the cutoff (Section 3.2).
- The local linear estimator at the MSE-optimal bandwidth has a bias of 0.0189, which is 47 percent of the effect, and coverage of 0.87 with conventional standard errors. The bias is not a mistake: the bandwidth that minimizes the mean squared error balances bias against variance, and the conventional interval ignores the bias. Section 4 is about that gap.
- Halving the bandwidth cuts the bias to 0.0053 and raises the standard deviation to 0.0547; the local quadratic at the full bandwidth does the same trade in a different way. Bias and variance are the currency of every choice in this lecture.

1.5 Roadmap

Section 2 states what identifies the jump and what threatens it. Section 3 builds the local polynomial estimator and its bandwidth. Section 4 repairs the interval, with bias correction and with honest intervals. Section 5 is the diagnostic battery every RD paper reports. Section 6 brings in covariates and machine

learning. Section 7 is local randomization, the other way to think about the design. Section 8 goes beyond one sharp cutoff, Section 9 to kinks and bunching, and Section 10 to the replications and a checklist. Throughout, the estimators are those of the `rdrobust` family and the package adds the pieces around them.

2 Identification

2.1 Sharp RD: continuity at the cutoff

Write $\mu_d(x) = E[Y(d) \mid X = x]$. The observed conditional mean is $E[Y \mid X = x] = \mu_1(x)$ for $x \geq c$ and $\mu_0(x)$ for $x < c$.

Assumption (continuity). μ_1 and μ_0 are continuous at c , and the density of X is positive at c (Hahn, Todd, and van der Klaauw 2001).

Then

$$\lim_{x \downarrow c} E[Y \mid X = x] - \lim_{x \uparrow c} E[Y \mid X = x] = \mu_1(c) - \mu_0(c) = \tau_{RD}.$$

Skeleton. The right limit is a limit of $\mu_1(x)$ and the left limit a limit of $\mu_0(x)$; continuity turns each into its value at c . Positive density makes both limits estimable. Appendix 11.1 writes this out with the potential-outcome notation.

Two remarks. Continuity is an assumption about the *potential* outcome functions, which no data can verify at c itself; its testable implications are the balance and density checks of Section 5. And nothing requires the density of X to be continuous at c for identification; a density jump is evidence about manipulation, which threatens continuity of μ_d , not a requirement in itself.

2.2 Local randomization as the other assumption

A different story: inside a small window $[c-w, c+w]$, being above or below the cutoff is as good as randomly assigned, and the running variable has no effect on the outcome within the window (Cattaneo, Frandsen, and Titiunik 2015). This is stronger than continuity: it asks the potential outcome functions to be flat, not merely continuous, inside the window. It buys exact finite-sample inference by permutation and handles discrete running variables and small samples where local polynomials cannot be fitted. Section 7 uses it.

2.3 Fuzzy RD: the cutoff as an instrument

When the cutoff changes the probability of treatment without determining it, $P(D = 1 \mid X = x)$ jumps at c from p^- to $p^+ < 1$. The jump in the outcome divided by the jump in treatment,

$$\tau_{FRD} = \frac{\lim_{x \downarrow c} E[Y \mid X = x] - \lim_{x \uparrow c} E[Y \mid X = x]}{\lim_{x \downarrow c} E[D \mid X = x] - \lim_{x \uparrow c} E[D \mid X = x]},$$

is the effect for compliers at the cutoff, the units whose treatment is switched by crossing it, under the monotonicity and exclusion conditions of Section 05 applied locally. The indicator $1\{X \geq c\}$ is the instrument, the running variable and its interaction with the indicator are the controls, and everything Section 05 said about weak instruments applies inside the window (Section 3.6).

2.4 Threats

Three things break continuity of μ_d at c .

- **Manipulation.** If units can place themselves on the preferred side of the cutoff, those just above differ from those just below in whatever drove the manipulation. Precise control over X is the problem; partial control with an idiosyncratic component is not (Lee 2008), because the units who end up near the cutoff are still a mix. The density test looks for the pile-up that precise control creates.
- **Other discontinuities at the cutoff.** If a second policy uses the same cutoff, the jump combines both. The remedy is institutional knowledge, not statistics.

- **Heaping.** Running variables recorded in round numbers (birth weight in 100-gram multiples, grades on a coarse scale) put atypical units at the heaps; if a heap sits at the cutoff, the comparison at the cutoff is between atypical groups (Barreca et al. 2011). Donut-hole estimates and discrete-score methods respond to this.

Covariate jumps at the cutoff are the symptom of all three, which is why balance is checked the way it is checked in an experiment.

3 Estimation

3.1 Three approximations of one target

The target is a difference of two limits of a conditional mean. Any consistent estimator of $E[Y | X = x]$ evaluated on each side and pushed to c will do; the estimators differ in how they trade bias for variance.

- **Local means.** Average Y in $[c - h, c)$ and in $[c, c + h]$ and subtract. Bias of order h from the slope; variance of order $1/(nh)$.
- **Global polynomials.** Fit Y on a polynomial in $X - c$ and its interaction with D on the whole support. Uses all data; the estimate at c depends on the shape far away, and no criterion chooses the order.
- **Local polynomials.** Fit a polynomial of low order p inside a bandwidth h with kernel weights, separately on each side, and take the difference of the intercepts. Bias of order h^{p+1} ; variance of order $1/(nh)$. With $p = 1$ this is the local linear estimator and the default of every RD package.

3.2 The local linear estimator as weighted least squares

On the right side, with K a kernel and h a bandwidth,

$$(\hat{\alpha}_+, \hat{\beta}_+) = \arg \min_{\alpha, \beta} \sum_{i: X_i \geq c} K\left(\frac{X_i - c}{h}\right) (Y_i - \alpha - \beta(X_i - c))^2,$$

and likewise on the left; $\hat{\tau} = \hat{\alpha}_+ - \hat{\alpha}_-$. Because weighted least squares is linear in Y , the estimate is a weighted sum $\hat{\tau} = \sum_i w_i(X_i; h) Y_i$ with weights that depend on the running variable only. The weights are the estimator's identity card.

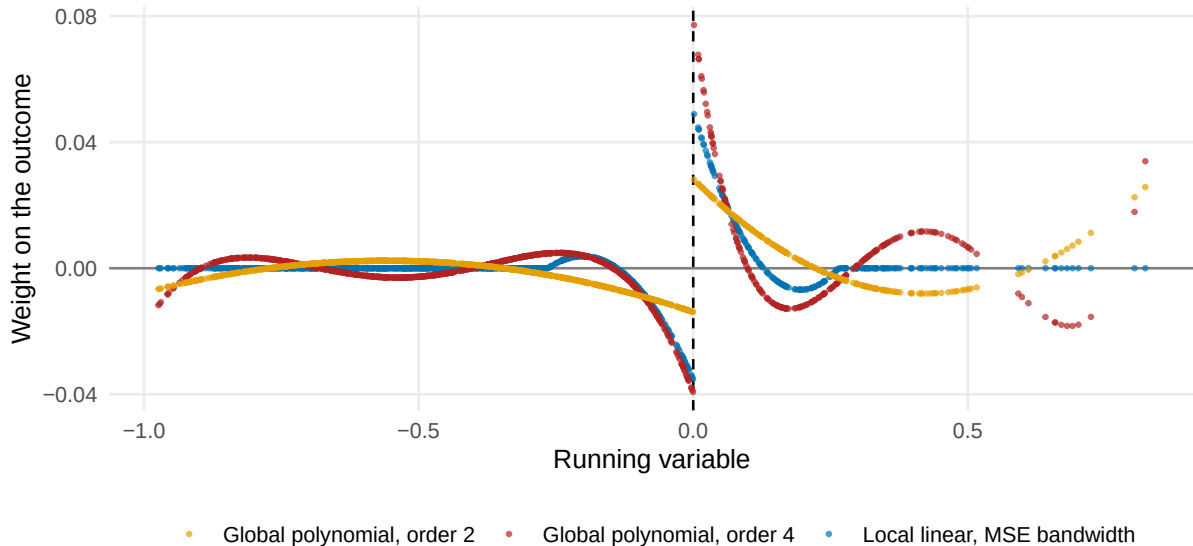


Figure 3: Weights on each observation's outcome in three estimators of the jump on the same sample of 1,000 units. The local linear estimator (triangular kernel, MSE-optimal bandwidth) uses only nearby units and its weights fall to zero at the bandwidth. The global quartic gives units far from the cutoff weights comparable to nearby units, some with the opposite sign.

3.3 Bias and variance

Under twice-differentiable conditional means and $h \rightarrow 0$ with $nh \rightarrow \infty$,

$$\hat{\tau} \approx N\left(\tau + h^2 B, \frac{V}{nh}\right), \quad B = \frac{\bar{\nu}}{2}(\mu_1''(c) - \mu_0''(c)), \quad V = \frac{\bar{\kappa}}{f_X(c)}(\sigma_1^2(c) + \sigma_0^2(c)),$$

with kernel constants $\bar{\nu}, \bar{\kappa}$ and conditional variances σ_d^2 . *Skeleton*. Expand $\mu_d(x)$ to second order around c ; the local linear fit is exact for the linear part, so the bias is the curvature term times the second moment of the kernel weights, which scales with h^2 . The effective sample size is proportional to nh , so the variance scales with $1/(nh)$. Appendix 11.2 gives the argument and the minimax rate $n^{-4/5}$ for this class.

Minimizing $h^4 B^2 + V/(nh)$ gives the MSE-optimal bandwidth $h^* \propto n^{-1/5}$. Two consequences shape everything below: at h^* the squared bias and the variance are of the same order, so the bias does not vanish relative to the standard error; and h^* depends on B and V , which must themselves be estimated from the data.

3.4 Kernel and order

The triangular kernel is optimal for estimating a boundary point among kernels, and it is the default. Local linear ($p = 1$) is preferred to local constant because the boundary bias of the constant fit is of order h ; higher orders reduce bias at the cost of variance and are the standard choice for the bias correction step ($q = p + 1$), not for the point estimate. The experiment of Section 1.4 showed the local quadratic at the linear bandwidth: less bias, more spread.

3.5 Bandwidth selection

Imbens and Kalyanaraman (2012) plug estimates of B and V into h^* . Calonico, Cattaneo, and Titiunik (2014) refine the pilot estimates and add a second bandwidth b for the bias correction of Section 4.2. Calonico, Cattaneo, and Farrell (2020) derive the bandwidth that minimizes the coverage error of the interval rather than the MSE of the point estimate, which is smaller than h^* by a factor $n^{-1/20}$. `rdbwselect()` returns all of them.

```
dat <- sim_rd(3000, "lee", seed = 1)
bw <- rdwbselect(dat$y, dat$x, c = 0, all = TRUE)
bw$bws
#>           h (left) h (right)  b (left) b (right)
#> mserd      0.1853957 0.1853957 0.3170565 0.3170565
#> msetwo      0.1692286 0.1565937 0.2654035 0.2442734
#> msesum      0.1623025 0.1623025 0.2538229 0.2538229
#> msecmb1     0.1623025 0.1623025 0.2538229 0.2538229
#> msecmb2     0.1692286 0.1623025 0.2654035 0.2538229
#> cerrd      0.1242349 0.1242349 0.3170565 0.3170565
#> certwo      0.1134012 0.1049345 0.2654035 0.2442734
#> cersum      0.1087600 0.1087600 0.2538229 0.2538229
#> cercomb1    0.1087600 0.1087600 0.2538229 0.2538229
#> cercomb2    0.1134012 0.1087600 0.2654035 0.2538229
```

The rows differ in whether the two sides share a bandwidth (`mserd`, `cerrd`) or not (`msetwo`, `certwo`) and in the criterion (MSE or coverage error). The reporting norm is to show the MSE-optimal choice and a sensitivity figure over a grid (Section 5); a result that survives only at one bandwidth is not a result.

3.6 Fuzzy RD estimation

Two local linear fits and a ratio: the jump in Y over the jump in D , both at the same bandwidth. Equivalently, local two-stage least squares of Y on D with $1\{X \geq c\}$ as instrument and $(X - c)$, $1\{X \geq c\}(X - c)$ as controls, weighted by the kernel. `rdrobust(fuzzy = )` does the former with bias correction of the ratio.

```
fz <- sim_rd(3000, "fuzzy", compliance = 0.8, seed = 2)
summary(rdrobust(fz$y, fz$x, c = 0, fuzzy = fz$d))
#> Call: rdrobust
```

```

#>
#> Fuzzy RD estimates using local polynomial regression.
#>
#> Number of Obs.          3000
#> BW type                mserd
#> Kernel                  Triangular
#> VCE method              NN
#>
#>                        Left      Right
#> Number of Obs.        2404      596
#> Eff. Number of Obs.   518      328
#> Order est. (p)         1         1
#> Order bias (q)         2         2
#> BW est. (h)            0.216    0.216
#> BW bias (b)            0.340    0.340
#> rho (h/b)             0.635    0.635
#> Unique Obs.           2404      596
#>
#> First-stage estimates.
#>
#> =====
#>                Point   Robust Inference
#>                Estimate      z      P>|z|      [ 95% C.I. ]
#> =====
#>      Rd Effect    0.580    8.724    0.000    [0.451 , 0.712]
#> =====
#>
#> Treatment effect estimates.
#>
#> =====
#>                Point   Robust Inference
#>                Estimate      z      P>|z|      [ 95% C.I. ]
#> -----
#>      RD Effect    0.088    2.145    0.032    [0.007 , 0.162]
#> =====

```

The first-stage jump is 0.60 in this design, a strong instrument. When the jump is small, the ratio inherits the weak-instrument problem of Section 05: the Wald interval is too short and its coverage falls (Feir, Lemieux, and Marmar 2016). `rd_weak_iv()` computes the local first-stage F statistics and the Anderson-Rubin set on the kernel-weighted sample, which stays valid however small the jump.

```

weak <- sim_rd(3000, "fuzzy", compliance = 0.55, seed = 2)
wk <- rd_weak_iv(weak, "y", "d", "x")
wk
#> Fuzzy RD with weak-identification-robust inference (h = 0.2461, n in window = 945)
#> local first stage: F = 6.69, robust F = 4.18, effective F = 4.18
#> Anderson-Rubin set: bounded
#>
#>                method estimate std.error conf.low conf.high
#> local 2SLS (Wald)  0.2362    0.1593 -0.07608    0.5485
#> Anderson-Rubin    NA        NA    -0.01443    6.2698
#> rdrobust conventional 0.2362    0.1580 -0.07354    0.5460
#> rdrobust bias_corrected 0.1628    0.1580 -0.14693    0.4726
#> rdrobust robust    0.1628    0.2286 -0.28527    0.6109

```

With a first-stage jump of 0.1 the effective F is 4.2, far below the usual thresholds. The Wald interval ends at 0.55; the Anderson-Rubin set is bounded and reaches 6.27. The data cannot pin the ratio down, and the Wald interval's width is an artifact of treating a noisy denominator as fixed.

4 Inference

4.1 Why the MSE-optimal bandwidth breaks the naive interval

At h^* the bias $h^{*2}B$ and the standard error $\sqrt{V/(nh^*)}$ are both of order $n^{-2/5}$. The conventional interval $\hat{\tau} \pm 1.96 \widehat{se}$ is centered at a point that misses the truth by a fixed multiple of its width, so its coverage is

below nominal no matter how large n is. Section 1.4 measured that coverage at 0.87.

4.2 Undersmoothing, bias correction, and the robust variance

Two repairs. *Undersmoothing* picks h smaller than h^* so that the bias is negligible relative to the standard error; the point estimate is noisier, and no data-driven rule says how much smaller. *Bias correction* (Calonico, Cattaneo, and Titiunik 2014) keeps h^* , estimates B with a local quadratic fit at a second bandwidth b , and subtracts $h^2 \hat{B}$. The estimated bias has its own sampling variability, which the conventional standard error ignores; the *robust* standard error adds it. That is the three-row output of `rdrobust()`: conventional (point estimate, naive SE), bias-corrected (corrected point estimate, naive SE), robust (corrected point estimate, SE that accounts for the correction). Only the last row has asymptotically valid coverage at h^* ; in samples of a thousand the gap to nominal coverage closes slowly, as the experiment below shows. Appendix 11.4 states the result.

Table 2: Coverage of four 95 percent intervals on the Lee design, $n = 1,000$, 200 simulations. The first three come from `rdrobust()` at the MSE-optimal bandwidth; the fourth from `RDHonest()` with the rule-of-thumb smoothness bound.

Interval	Bias of the center	95% coverage	Mean length	Mean bandwidth
Conventional	0.0220	0.84	0.1428	0.1999
Bias-corrected	0.0155	0.83	0.1428	0.1999
Robust bias-corrected	0.0155	0.88	0.1676	0.1999
Honest (RDHonest)	0.0098	0.93	0.2106	0.1127

The conventional interval covers 0.84 of the time and the robust bias-corrected interval 0.88, at the cost of a length 1.17 times larger. The bias-corrected row without the robust variance shows why both steps are needed: recentering alone leaves coverage at 0.83. The correction removes only part of the bias at this sample size (the center’s bias falls from 0.0220 to 0.0155), because the curvature is itself estimated with error; the robust variance accounts for that error but cannot remove it, which is why the honest interval of the next section gets closer to nominal coverage.

4.3 Honest inference

Bias correction estimates the curvature. Armstrong and Kolesár (2018, 2020) start from a different question: if all we assume is that the second derivative of the conditional mean is bounded by M , what is the shortest interval with guaranteed coverage? The worst-case bias of any linear estimator with weights w_i is $\frac{M}{2} \sum_i |w_i| (X_i - c)^2$, a number computable from the weights alone; the honest interval is $\hat{\tau} \pm cv_\alpha(\text{bias}/se) \cdot se$ with a critical value from the non-central normal distribution, and the bandwidth is chosen to minimize its length. Appendix 11.5 has the calculation.

The motivation is a negative result: without a bound like M no interval can have finite expected length and correct coverage uniformly over the continuous functions the identification assumption allows (Bertanha and Moreira 2020), so any procedure that estimates the smoothness must undercover somewhere. The price of honesty is the choice of M ; `RDHonest` offers a rule of thumb from a global quartic and the analysis should be reported for a range of M .

```
hn <- RDHonest::RDHonest(y ~ x, data = dat, cutoff = 0)
tidy(hn)
#>      term      estimate std.error statistic  p.value  conf.low
#> 1 Sharp RD parameter 0.04828648 0.02500589  1.931005 0.1047357 -0.00943586
#>      conf.high
#> 1 0.1060088
```

The honest row of Table 2 covers 0.93 of the time with a length 1.26 times the robust bias-corrected interval; it also chooses a smaller bandwidth, because interval length rather than mean squared error is the criterion.

4.4 Discrete running variables

When the running variable takes few values (grade point averages, ages in years, test scores in integers), the observations at the cutoff share the same X and the local polynomial's variance formula, which treats X as continuous, understates the uncertainty of the extrapolation between mass points (Kolesár and Rothe 2018). Clustering by X does not fix it. The honest interval does, because it never needs the density to be continuous: the bound on curvature controls the interpolation between the mass points nearest the cutoff.

```
dd <- sim_rd(3000, "discrete", grid = 0.1, seed = 3)
length(unique(dd$x))
#> [1] 20
rd_frame(rdrobust = rdrobust(dd$y, dd$x, c = 0), honest = RDHonest::RDHonest(y ~ x, data = dd, cutoff = 0),
  methods = "robust")
#>      model      term method estimate std.error  conf.low conf.high
#> 1 rdrobust      RD effect robust 0.10499059 0.02588256 0.0542617 0.1557195
#> 2 honest Sharp RD parameter honest 0.08887367 0.02069106 -0.1174382 0.2951855
#>      p.value      h      n_eff
#> 1 4.983031e-05 0.2942127 934.0000
#> 2 9.999722e-01 0.2000000 908.8244
```

This is the setting of Bleemer and Mehta (2022), whose running variable is an average of two course grades, and who report honest intervals as the robustness check for that reason.

4.5 Clustering and geographic designs

Standard errors cluster at the level at which the running variable is assigned or the outcome is correlated: a school, a district, a border segment. `rdrobust(cluster = )` handles it. Geographic designs use distance to a boundary as the running variable and cluster by boundary segment (Keele and Titunik 2015); Bagwe, Margitic, and Stashko (2026) is the worked example in Section 10.

5 Diagnostics

5.1 The density test

If units sort across the cutoff, the density of X jumps there. McCrary (2008) estimated the density on each side with a smoothed histogram and tested the log difference; Cattaneo, Jansson, and Ma (2020) replaced the histogram with a local polynomial density estimator that needs no pre-binning and has bias-corrected inference. `rddensity()` implements the latter.

```
man <- sim_rd(4000, "manipulated", manip = 0.5, seed = 4)
summary(rddensity::rddensity(man$x, c = 0))
#>
#> Manipulation testing using local polynomial density estimation.
#>
#> Number of obs =      4000
#> Model =          unrestricted
#> Kernel =         triangular
#> BW method =      estimated
#> VCE method =     jackknife
#>
#> c = 0
#>      Left of c      Right of c
#> Number of obs      3091      909
#> Eff. Number of obs  526      496
#> Order est. (p)      2        2
#> Order bias (q)      3        3
#> BW est. (h)         0.215     0.159
#>
#> Method      T      P > |T|
#> Robust      3.1539    0.0016
#>
#>
#> P-values of binomial tests (H0: p=0.5).
#>
```

```

#> Window Length / 2      <c      >=c      P>|T|
#> 0.015                  20       48      0.0009
#> 0.030                  40       96      0.0000
#> 0.045                  58      168      0.0000
#> 0.060                  74      211      0.0000
#> 0.075                  98      276      0.0000
#> 0.089                 114      333      0.0000
#> 0.104                 143      384      0.0000
#> 0.119                 187      420      0.0000
#> 0.134                 237      449      0.0000
#> 0.149                 285      475      0.0000

```

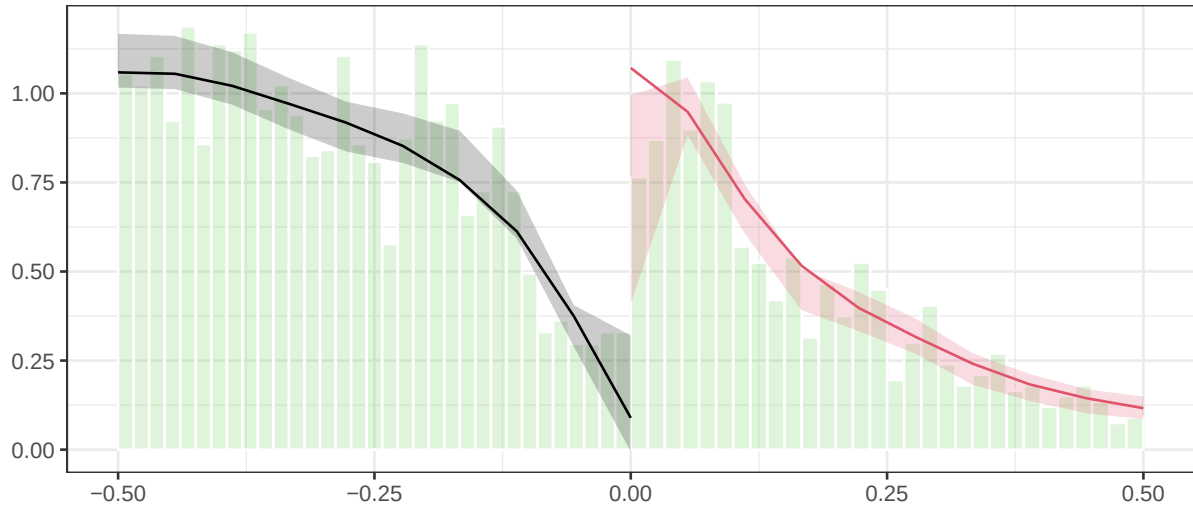


Figure 4: Density of the running variable in the manipulated design, where half of the units within 0.1 below the cutoff moved above it. Left: histogram. Right: the local polynomial density estimates on each side with confidence bands (rddensity).

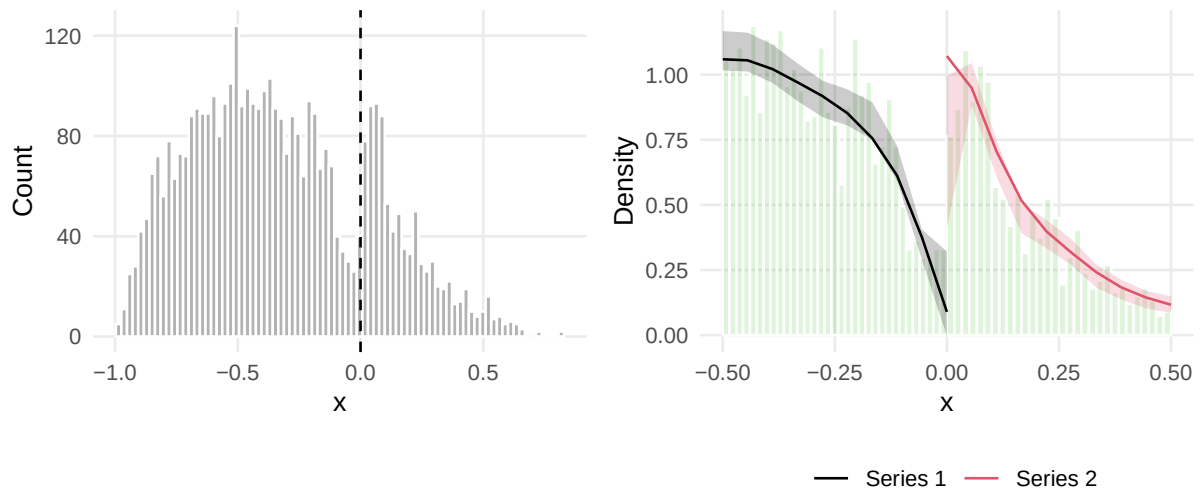


Figure 5: Density of the running variable in the manipulated design, where half of the units within 0.1 below the cutoff moved above it. Left: histogram. Right: the local polynomial density estimates on each side with confidence bands (rddensity).

A rejection means the density jumps, not that the design is invalid: the question is whether the movers differ in potential outcomes. Partial manipulation with an idiosyncratic element (a student who studies harder

but cannot control the exact score) leaves the comparison at the cutoff valid; precise control does not. A non-rejection is weak evidence, because sorting in both directions can cancel. The paper by Sejas-Portillo, Moro, and Stowasser (2025) shows both faces: sellers upgrade properties just below a label threshold, and the authors both test the density and remove the upgraded transactions.

5.2 Covariate balance

Predetermined covariates are placebo outcomes: the RD estimator applied to them should find nothing. `rd_balance()` runs `rdrobust()` on each covariate and adds a joint Wald test from the stacked influence functions of the local linear jumps at one common bandwidth, which the per-covariate output cannot give.

```
bal <- rd_balance(dat_c <- sim_rd(3000, "covariates", seed = 5), paste0("z", 1:4), "x")
bal
#> Covariate balance at the cutoff (robust bias-corrected rows)
#> covariate estimate std.error p.value h n_left n_right
#>      z1  0.03311    0.1917  0.8628 0.2306  533   342
#>      z2 -0.19821    0.1706  0.2453 0.2599  614   370
#>      z3 -0.02240    0.1960  0.9090 0.2257  522   337
#>      z4 -0.03438    0.1836  0.8515 0.2008  463   311
#> joint test of no jumps at h = 0.2282: chi2(4) = 1.63, p = 0.804
```

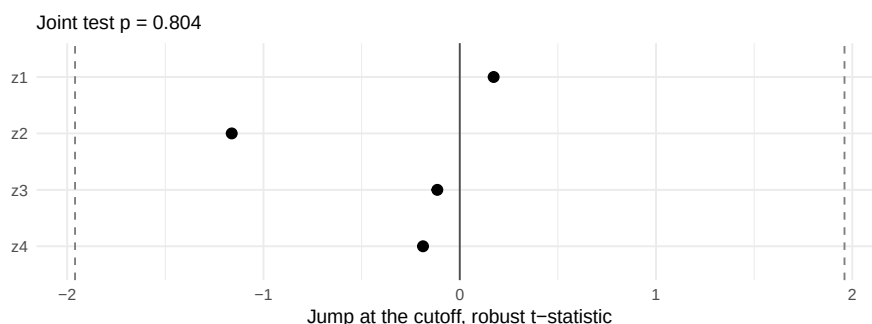


Figure 6: Robust t-statistics of the jump in each covariate at the cutoff (the layout of Lee 2008, Figures III and IV).

5.3 Placebo cutoffs, bandwidth sensitivity, donut holes

Three more checks, each a loop of estimates that a reader expects to see.

- **Placebo cutoffs.** Estimate the jump at points where no treatment changes, on each side using only that side's data so the true jump never enters (Imbens and Lemieux 2008). Significant placebo jumps mean the outcome function has features the local fit mistakes for jumps.
- **Bandwidth sensitivity.** Re-estimate over a grid around h^* , holding the ratio h/b fixed. Estimates that swing with h signal curvature the bias correction is not handling.
- **Donut holes.** Drop the observations within a radius of the cutoff and re-estimate. Estimates that change signal heaping or manipulation right at the threshold (Barreca et al. 2011).

`rd_checks()` runs all of them with the main estimate and the balance table, and `plot_rd_checks()` draws each.

```
ck <- rd_checks(dat_c, "y", "x", covariates = paste0("z", 1:4))
ck$main
#>      term      method estimate std.error statistic
#> Conventional RD effect conventional -0.09292455 0.08009507 -1.160178
#> Bias-Corrected RD effect bias_corrected -0.11747414 0.08009507 -1.466684
#> Robust      RD effect      robust -0.11747414 0.09369483 -1.253795
#>      p.value conf.low conf.high h_left h_right b_left
#> Conventional 0.2459763 -0.2499080 0.06405890 0.2493157 0.2493157 0.4110593
#> Bias-Corrected 0.1424621 -0.2744576 0.03950931 0.2493157 0.2493157 0.4110593
#> Robust      0.2099164 -0.3011126 0.06616435 0.2493157 0.2493157 0.4110593
```

```
#>           b_right n_left n_right
#> Conventional 0.4110593   593   361
#> Bias-Corrected 0.4110593   593   361
#> Robust       0.4110593   593   361
```

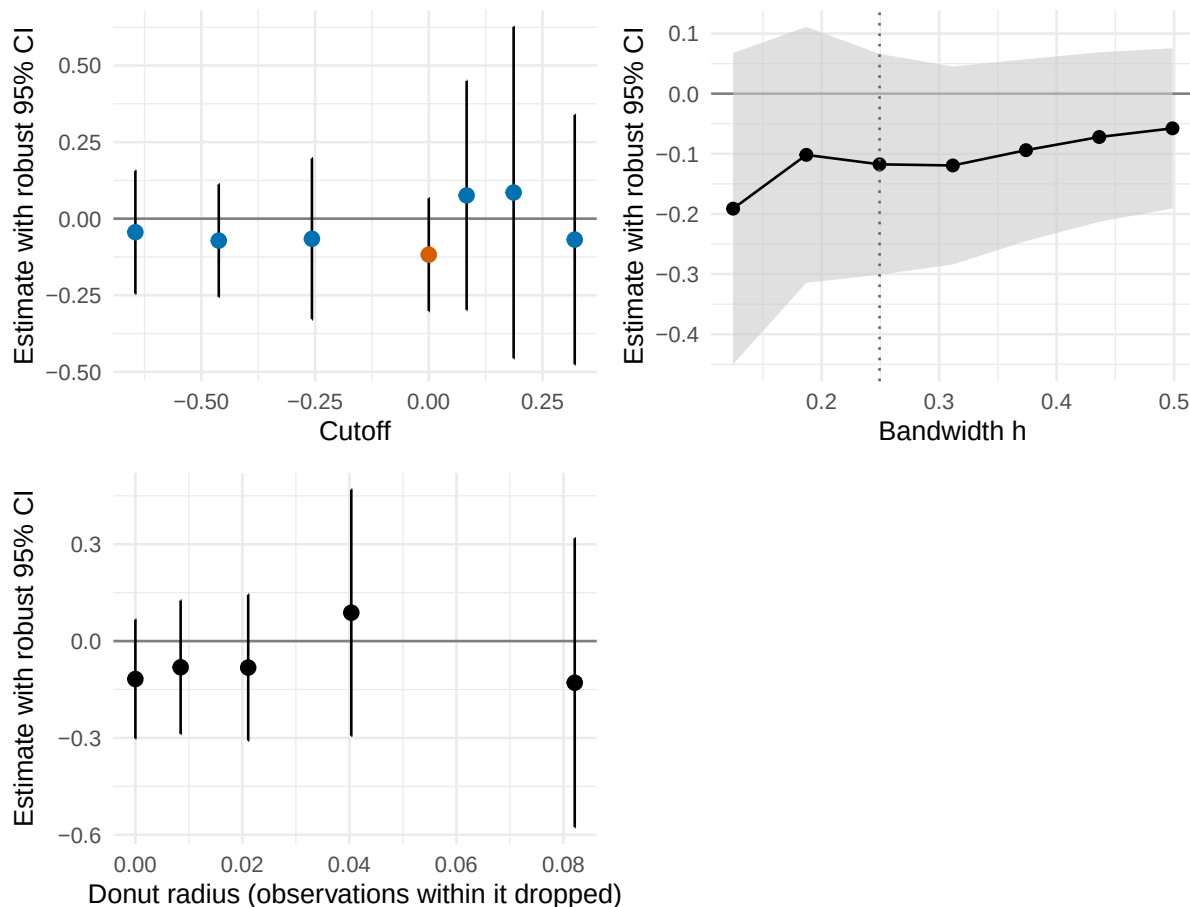


Figure 7: The diagnostic battery on the covariates design: placebo cutoffs (the true cutoff in orange), bandwidth sensitivity (dotted line at the MSE-optimal bandwidth), and donut-hole estimates. Robust 95 percent intervals throughout.

5.4 The RD plot done right

The plot is evidence, and it can mislead. Bins that are too fine show noise; bins that are too coarse hide a jump or invent one; a fitted polynomial can overshoot near the cutoff and make the fitted lines meet at a jump the bin means do not show. Calonico, Cattaneo, and Titiunik (2015) choose the number of bins to trade the bias of coarse bins against the variance of fine ones (the mimicking-variance choice makes the bin means as variable as the raw data, so the eye judges the jump against the right noise level). `rd_plot()` uses that choice by default and lets the fit be the estimator's own local linear lines rather than a global polynomial. Figure 1 showed both; report the local version when the estimate is local.

5.5 A diagnostics table for a paper

One table with the main estimate, the balance rows, the density test, the placebo cutoffs, and the donut estimates, plus the sensitivity figure, is what Lee (2008), Sejas-Portillo et al. (2025), and Bagwe et al. (2026) all report. `rd_frame()` assembles the estimate rows from any of the packages into one frame for that table.

```
rd_frame(main = ck$main_fit, honest = RDHonest::RDHonest(y ~ x, data = dat_c, cutoff = 0), methods = "robust")
#>      model      term method estimate std.error conf.low conf.high
#> 1  main      RD effect robust -0.1174741 0.09369483 -0.3011126 0.06616435
#> 2 honest Sharp RD parameter honest -0.1318233 0.10530967 -0.3582673 0.09462067
#>      p.value      h      n_eff
#> 1 0.2099164 0.2493157 954.0000
#> 2 0.2576887 0.1463864 464.4852
```

6 Covariates and machine learning

6.1 Covariates are not needed, and what they buy

Identification needs the running variable and nothing else. Covariates enter for three reasons: precision, because the outcome variation they explain is noise for the jump; heterogeneity, because the effect at the cutoff may differ across groups; and extrapolation, because covariates can carry the comparison away from the cutoff (Section 8). Two rules follow. Covariates must be predetermined, since a covariate that responds to treatment absorbs part of the effect. And the point estimate should not move much when they are added; if it does, the covariates jump at the cutoff and the design has a problem that precision cannot fix.

6.2 Linear adjustment inside the local regression

Calonico, Cattaneo, Farrell, and Titiunik (2019) add the covariates linearly to the local regression with a common coefficient on both sides. The result is the local linear estimator applied to the adjusted outcome $Y - Z'\hat{\gamma}$, with the same bias as without covariates and a smaller variance: the conditional variance of $Y - Z'\gamma_0$ replaces that of Y . *Skeleton*. The coefficient $\hat{\gamma}$ converges to a fixed vector; $E[Z \mid X = x]$ is smooth at c , so subtracting $Z'\gamma_0$ changes the two limits by the same amount and the jump is unchanged; what remains is the local linear estimator on a less noisy outcome. Appendix 11.3 states the result. `rdrobust(covs = )` implements it.

6.3 Flexible adjustment: subtract any prediction from the covariates

The skeleton did not use linearity. Any function $\eta(Z)$ with $E[\eta(Z) \mid X = x]$ continuous at c leaves the jump unchanged, and $Y - \eta(Z)$ has a smaller variance whenever η predicts Y . Noack, Olma, and Rothe (2024) turn this into a two-step procedure: estimate η with any machine learner, cross-fitted so that no observation's outcome enters its own adjustment, then run the standard local polynomial estimator on $Y_i - \hat{\eta}(Z_i)$. Because the estimand does not depend on η , the learner needs no rate condition for consistency, which is weaker than what Section 04 required for double machine learning: there the nuisances entered the estimand, here they only reduce noise. The recommended $\hat{\eta}$ is the average of the two side-specific predictions, which stays a function of Z alone. Kreiss and Rothe (2023) obtain the same structure with a local lasso when the covariates are many.

`rd_adjust()` implements the cross-fitted adjustment with any `mlr3` learner and, when `rdrobust` is installed, reports the raw and adjusted fits side by side.

```
z4 <- paste0("z", 1:4)
adj_lin <- rd_adjust(dat_c, "y", "x", covariates = z4, seed = 1)
adj_rf <- rd_adjust(dat_c, "y", "x", covariates = z4, pooled = FALSE, seed = 1,
  learner = lrn("regr.ranger", num.trees = 300, min.node.size = 20))
rd_frame(no_covariates = adj_lin$fits$raw,
  rdrobust_covs = rdrobust(dat_c$y, dat_c$x, c = 0, covs = as.matrix(dat_c[, z4])),
  linear_adjustment = adj_lin$fits$adjusted,
  forest_adjustment = adj_rf$fits$adjusted, methods = "robust")
#>      model      term method estimate std.error conf.low
#> 1  no_covariates RD effect robust -0.1174741375 0.09369483 -0.30111263
#> 2  rdrobust_covs RD effect robust -0.1091515544 0.08039520 -0.26672325
#> 3  linear_adjustment RD effect robust -0.1176066994 0.08054459 -0.27547119
#> 4  forest_adjustment RD effect robust -0.0007008075 0.03710393 -0.07342317
#>      conf.high p.value      h      n_eff
#> 1 0.06616435 0.2099164 0.2493157 954
```

```
#> 2 0.04842014 0.1745629 0.2166062 823
#> 3 0.04025779 0.1442505 0.2257588 859
#> 4 0.07202156 0.9849307 0.2013991 775
```

The design’s covariates enter the outcome nonlinearly (a quadratic in z_2). The linear adjustments, inside `rdrobust` or through `rd_adjust()`, remove the linear part and lower the standard error; the forest removes the curvature too and lowers it further. The point estimates move by less than two standard errors of the least precise one, as they should when the covariates are balanced; what changes is the precision.

6.4 Experiment: precision gains from adjustment

Table 3: Robust bias-corrected estimates with and without covariate adjustment on the covariates design, true effect 0.2, $n = 2,000$, 60 simulations (the layout of Table 17.1 in Chernozhukov et al. 2026).

Adjustment	Bias	SD	Mean robust SE	95% coverage
No covariates	0.0298	0.1185	0.1272	0.97
Linear (rdrobust covs)	0.0245	0.1115	0.1042	0.97
Linear adjustment (rd_adjust)	0.0282	0.1169	0.1070	0.98
Forest adjustment (rd_adjust)	0.0090	0.0617	0.0503	0.92

The forest adjustment cuts the standard deviation of the estimate from 0.1185 to 0.0617, a factor of 1.9, with no change in bias and coverage of 0.92. In sample-size terms, that factor squared is the multiple of observations the unadjusted estimator would need. Linear adjustment, through `rdrobust` or `rd_adjust()`, gets the part of the gain that a linear model can explain.

6.5 Heterogeneity at the cutoff

The effect at the cutoff may differ by group. The local linear regression with group indicators interacted with the treatment and the running variable estimates a group-specific jump, and a covariate dictionary in place of indicators gives the best linear predictor of the effect at the cutoff, with robust bias-corrected inference for each coefficient (Calonico, Cattaneo, Farrell, Palomba, and Titiunik 2025). This is the RD version of Section 08’s group effects and best linear predictors, and `rdhte` implements it.

```
dat_c$group <- as.integer(dat_c$z3 > 0)
ht <- rdhte::rdhte(dat_c$y, dat_c$x, c = 0, covs.hte = dat_c$group)
tidy(ht)
#>   term      estimate std.error statistic    p.value   conf.low  conf.high
#> 1    0 -0.18081837 0.2002598 -1.7342627 0.08287148 -0.7398051 0.04519888
#> 2    1  0.00618008 0.1413139  0.3914252 0.69548300 -0.2216563 0.33228397
#>   estimate.bc  h.left  h.right n.eff.left n.eff.right
#> 1 -0.34730309 0.2478763 0.2478763      303      189
#> 2  0.05531382 0.2344204 0.2344204      266      165
```

In this design the effect does not vary with z_3 , and the two group estimates agree within their standard errors.

7 Local randomization

7.1 The assumption and its window

Inside a window $W = [c - w, c + w]$, assume that treatment is as good as randomly assigned and that the potential outcomes do not depend on the running variable. Then the window is an experiment: differences in means estimate the effect for units in the window, and Fisher’s randomization test gives exact finite-sample p-values by permuting treatment among the units in W . The estimand is an average over the window, not the effect at the point c .

The window is the tuning parameter, in the role the bandwidth played before. Cattaneo, Frandsen, and Titiunik (2015) choose the largest window in which predetermined covariates are balanced, testing balance in nested windows and stopping when a test rejects.

```
win <- rdlocrand::rdwinselect(dat_c$x, as.matrix(dat_c[, c("z2", "z3", "z4")]), quietly = TRUE,
                             wmin = 0.02, wstep = 0.02, nwindows = 15)
c(window_left = win$w_left, window_right = win$w_right)
#> window_left window_right
#> -0.14 0.14
lr <- rdlocrand::rdrandinf(dat_c$y, dat_c$x, wl = win$w_left, wr = win$w_right, quietly = TRUE, ci = 0.05)
rd_frame(continuity = rdrobust(dat_c$y, dat_c$x, c = 0), local_randomization = lr, methods = "robust")
#> model term method estimate std.error
#> 1 continuity RD effect robust -0.11747414 0.09369483
#> 2 local_randomization RD effect local_randomization 0.07042247 NA
#> conf.low conf.high p.value h n_eff
#> 1 -0.3011126 0.06616435 0.2099164 0.2493157 954
#> 2 -0.0100000 0.17000000 0.1840000 0.1400000 3000
```

The window is selected on z_2 to z_4 only, starting from a half-width of 0.02 and growing in steps of 0.02 (the default starts from the ten observations nearest the cutoff, where a balance test at the 0.15 level rejects by chance often). The design's z_1 is correlated with the running variable, so it differs across the two sides of any window and no window passes its balance test: local randomization requires covariates that are balanced in a neighborhood, not merely continuous at the cutoff, which is the flatness assumption at work.

7.2 When to prefer it

Local randomization is the right tool when the local polynomial cannot be fitted: few observations near the cutoff, a running variable with a handful of values, or an outcome so noisy that any bandwidth large enough to estimate a slope is too large to trust. It is the wrong tool when the outcome moves with the running variable inside any usable window, because the flatness assumption then fails; the balance-based window selection protects against the covariates but not against the outcome itself. Reporting both estimates, as above, is the usual practice: agreement across two different assumptions is stronger evidence than either alone.

8 Beyond the single sharp cutoff

8.1 Multiple cutoffs

Many rules use different cutoffs for different units: energy-label thresholds at six scores, scholarship cutoffs that differ by year or district. Two estimands follow. Cutoff-specific effects treat each cutoff as its own design. The normalize-and-pool estimate recenters every unit's score at its own cutoff and runs one RD; it estimates a weighted average of the cutoff-specific effects with weights proportional to the density of units at each cutoff (Cattaneo, Keele, Titiunik, and Vazquez-Bare 2016), and it is the right summary only when the effects are similar or the weights are the ones a decision maker wants. `rdmulti::rdmc()` reports both, and the difference between cutoffs is itself evidence about extrapolation: with a common untreated function across cutoffs one can predict the effect at one cutoff for units assigned to another (Cattaneo et al. 2021).

```
md <- sim_rd(4000, "multi_cutoff", seed = 6)
attr(md, "tau_true")
#> [1] 0.04 0.07 0.10
mc <- rdmulti::rdmc(md$y, md$x, C = md$cutoff, verbose = FALSE, plot = FALSE)
#>
#> Cutoff-specific RD estimation with robust bias-corrected inference
#> =====
#> Cutoff Coef. P-value 95% CI hl hr Nh Weight
#> =====
#> -0.500 0.098 0.004 0.034 0.188 0.113 0.113 305 0.291
#> 0.000 0.093 0.008 0.026 0.174 0.119 0.119 341 0.325
#> 0.500 0.060 0.096 -0.010 0.123 0.144 0.144 402 0.384
#> -----
#> Weighted 0.082 0.000 0.045 0.128 . . 1048 .
```

```

#> Pooled      0.085 0.000    0.049    0.131    0.123    0.123    1014      .
#> =====
rd_frame(multi = mc)
#>   model      term method  estimate std.error  conf.low conf.high
#> 1 multi cutoff 1 robust 0.09833108 0.03906002  0.03448868 0.1876012
#> 2 multi cutoff 2 robust 0.09286118 0.03770790  0.02573728 0.1735495
#> 3 multi cutoff 3 robust 0.06000113 0.03389312 -0.01008728 0.1227713
#> 4 multi weighted robust 0.08184837 0.02118464  0.04483058 0.1278728
#> 5 multi pooled robust 0.08472038 0.02105648  0.04873723 0.1312771
#>      p.value      h n_eff
#> 1 4.470085e-03 0.1133538 179
#> 2 8.229466e-03 0.1185074 209
#> 3 9.644391e-02 0.1441712 248
#> 4 4.578806e-05      NA 636
#> 5 1.915164e-05 0.1226201 605

```

Sejas-Portillo, Moro, and Stowasser (2025) estimate threshold by threshold, because the value of moving from label C to B need not equal that of moving from F to E, and they read the pattern across thresholds as the result.

8.2 Multi-score and geographic designs

When two scores define treatment (two test cutoffs, or latitude and longitude on either side of a boundary), the cutoff is a curve and the effect is a function along it. Estimation picks boundary points and runs an RD on the distance to each point, or uses the signed distance to the boundary as a single score (Keele and Titiunik 2015). The single-score version assumes the effect is constant along the boundary; clustering by boundary segment handles the dependence of nearby units. Bagwe, Margitic, and Stashko (2026) use the signed distance to the precinct border with the polling-place distance as a continuous treatment, a fuzzy design whose first stage is the jump in distance at the border; `rdmulti::rdms()` implements the multi-score version.

8.3 Difference-in-discontinuities and RD in time

When the same cutoff existed before the policy, the jump before the policy measures the other discontinuities at the cutoff, and the change in the jump is the effect (Grembi, Nannicini, and Troiano 2016). With `fixest` this is the local regression with a period indicator interacted with the jump. RD in time uses the date as the running variable; it inherits every problem of event studies (anticipation, serial correlation, other events) and Hausman and Rapson (2018) list the additional checks it needs.

8.4 Extrapolation with covariates

The effect at the cutoff is the RD estimand; effects for units away from it need an assumption that the design does not provide. Angrist and Rokkanen (2015) assume that, within a window, potential outcomes are mean independent of the running variable given covariates: the covariates carry all the information the score has about outcomes. The assumption has a testable implication, that the score does not predict the outcome given the covariates on either side, and it identifies the untreated mean of the treated side as the covariate-adjusted mean of the untreated side. `rd_extrapolate()` runs the test and the extrapolation, by regression or doubly robustly with the side indicator as the treatment, and traces the effect along the running variable.

```

cia <- sim_rd(4000, "cia", seed = 7)
ex <- rd_extrapolate(cia, "y", "x", covariates = "z", window = 0.5)
ex
#> RD extrapolation under conditional independence (regression), window = 0.5, n = 2488
#>
#> CIA test (running-variable terms given covariates, per side):
#>   side    n coef_x   se_x statistic df p.value
#> below 1855 0.08559 0.09119    1.5290  2  0.4656
#> above  633 0.08658 0.14449    0.3591  2  0.8356
#>

```

```
#> Average effects away from the cutoff:
#>   side estimate std.error      n conf.low conf.high
#> above  0.4886 0.006761   633  0.4753  0.5018
#> below  0.4856 0.007782  1855  0.4703  0.5008
```

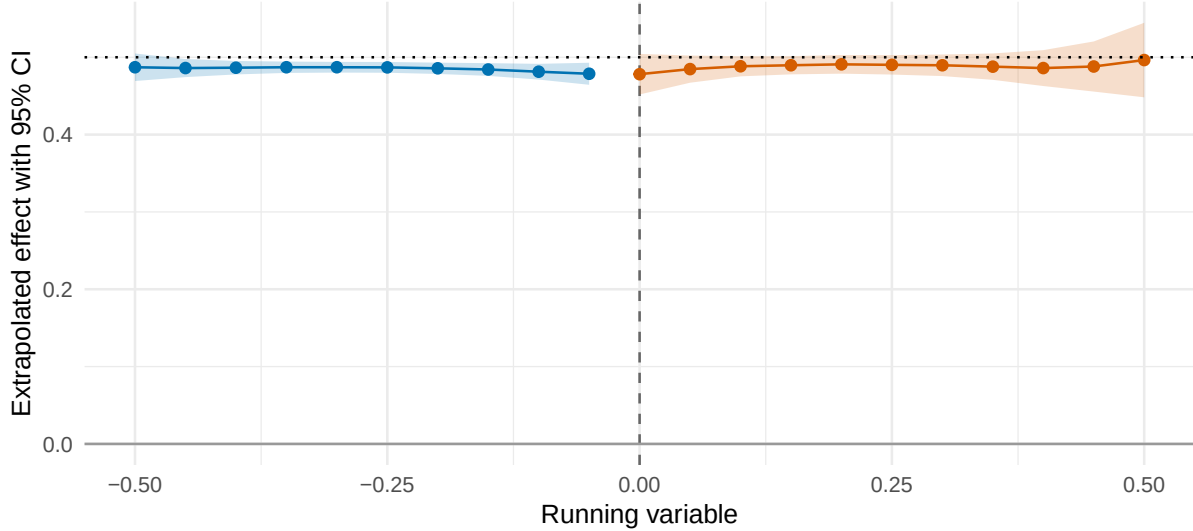


Figure 8: Extrapolated effects along the running variable in a design where the outcome depends on the score only through the covariate. The dotted line is the true effect, constant at 0.5.

The test does not reject on either side and the effects away from the cutoff recover the truth. On the Lee design, where the outcome depends on the score directly, the same test rejects at once:

```
bad <- sim_rd(4000, "lee", seed = 7)
bad$z <- rnorm(4000)
rd_extrapolate(bad, "y", "x", covariates = "z", window = 0.5)$test
#>   side      n   coef_x      se_x statistic df      p.value
#> 1 below 1855 0.2440559 0.09160851 357.91632  2 1.903084e-78
#> 2 above  633 0.5570856 0.14499348  99.55879  2 2.404820e-22
```

9 Regression kink and bunching

9.1 From jumps to kinks

Some rules change the slope of the treatment rather than its level: benefits replace 55 percent of earnings up to a cap, so the benefit's slope in earnings drops from 0.55 to zero at the cap. A kink in the treatment produces a kink in the outcome, and the regression kink design estimates the ratio of the two slope changes (Card, Lee, Pei, and Weber 2015):

$$\tau_{RKD} = \frac{\lim_{x \downarrow c} \partial_x E[Y | X = x] - \lim_{x \uparrow c} \partial_x E[Y | X = x]}{\lim_{x \downarrow c} \partial_x E[D | X = x] - \lim_{x \uparrow c} \partial_x E[D | X = x]}.$$

In a sharp kink the denominator is known from the rule; in a fuzzy kink it is estimated. The assumptions mirror the RD ones one derivative up: the density of the score and the conditional means of covariates must be smooth (differentiable) at the kink, and units must not sort at it, with the same allowance for imprecise control.

9.2 Estimation and why kinks are hard

A slope change is a derivative, and derivatives are estimated with more variance than levels: the bias of the local linear slope is of order h and its variance of order $1/(nh^3)$, so the optimal bandwidth is wider and

the estimate noisier than for a jump at the same n . `rdrobust(deriv = 1)` selects the bandwidth for the derivative; `rd_kink()` reports the reduced form, the first stage, the ratio, and the elasticity.

```
kd <- sim_rd(4000, "kink", seed = 8)
kd$z1 <- kd$x + rnorm(4000)
kk <- rd_kink(kd, "y", "x", d = "d", elasticity = TRUE)
kk$kink
```

	term	method	estimate	std.error	statistic
#> Conventional	kink ratio	conventional	1.535456	0.3410765	4.501794
#> Bias-Corrected	kink ratio	bias_corrected	1.266686	0.3410765	3.713789
#> Robust	kink ratio	robust	1.266686	0.5014163	2.526217

	p.value	conf.low	conf.high	h_left	h_right	b_left
#> Conventional	6.738217e-06	0.8669586	2.203954	0.1303471	0.1303471	0.2741236
#> Bias-Corrected	2.041787e-04	0.5981886	1.935184	0.1303471	0.1303471	0.2741236
#> Robust	1.152984e-02	0.2839283	2.249444	0.1303471	0.1303471	0.2741236

	b_right	n_left	n_right
#> Conventional	0.2741236	353	288
#> Bias-Corrected	0.2741236	353	288
#> Robust	0.2741236	353	288

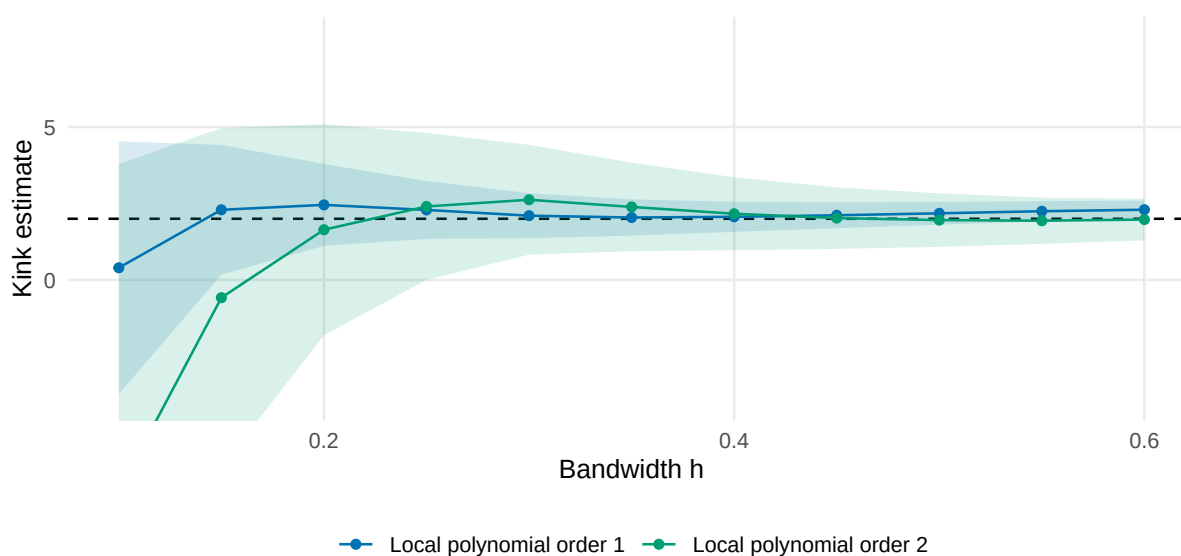


Figure 9: Fuzzy kink estimates with robust 95 percent intervals across bandwidths for local linear and local quadratic fits on one sample of 4,000 units (the pattern of Card et al. 2015, Figure 4). The dashed line is the true ratio.

The figure reproduces the lesson of Card et al.: local linear estimates settle at bandwidths several times the RD-typical ones, and local quadratic estimates are too imprecise to be informative at any bandwidth the sample can support. Kink designs need many observations and a transparent bandwidth choice; a kink estimate reported at one narrow bandwidth should be read with suspicion.

9.3 The four kink graphs

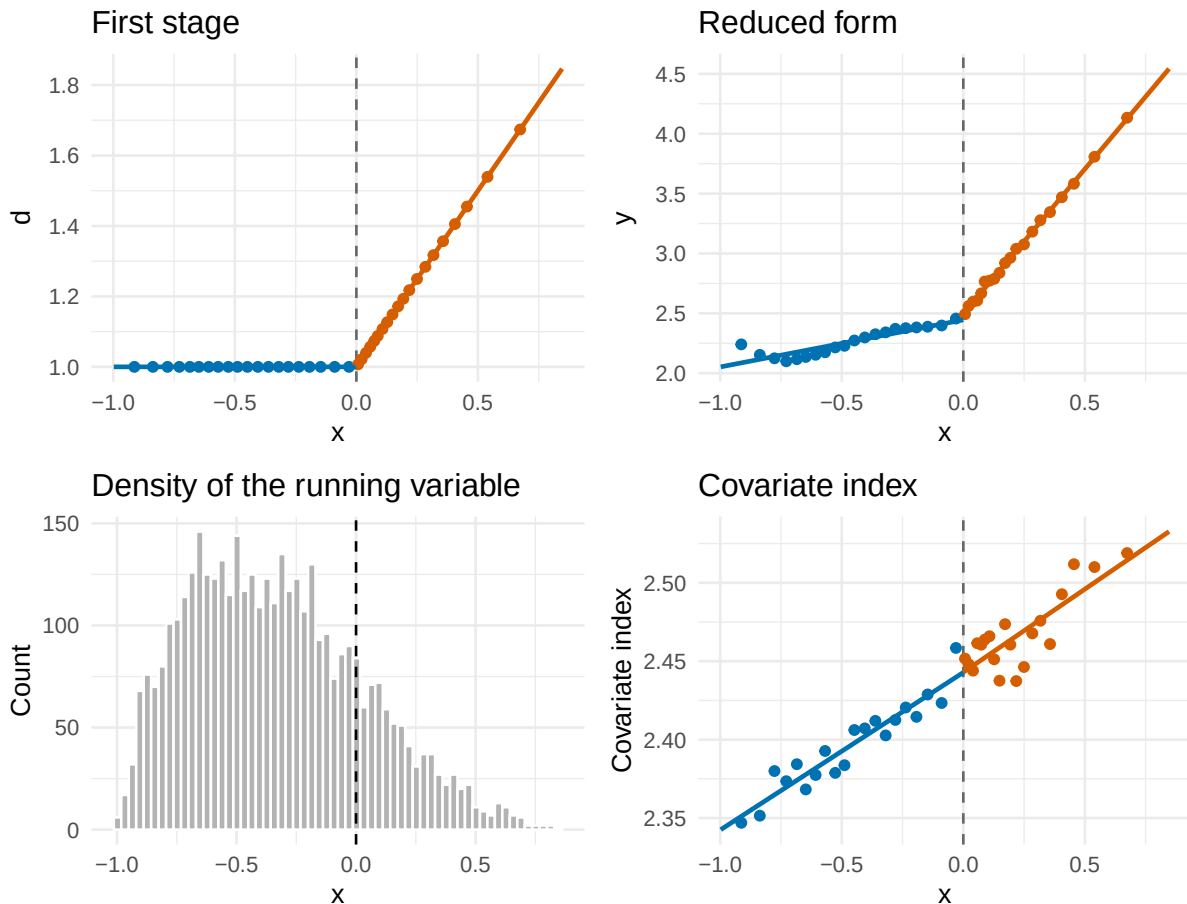


Figure 10: The four regression kink graphs: first stage, reduced form, density of the running variable, and a covariate index, each with binned means and linear fits per side.

The first stage shows the kink in the rule, the reduced form the kink in the outcome, and the last two show that nothing else kinks there. Landais (2015) reports exactly these four for the unemployment benefit schedule in Section 10.

9.4 Bunching: when manipulation is the object

Section 5.1 treated a density jump as a threat. When the rule creates an incentive to be on one side, the mass that piles up at the kink or notch measures the behavioral response: the excess mass relative to a smooth counterfactual density, divided by the change in the incentive, is an elasticity (Saez 2010; Chetty et al. 2011; Kleven 2016). The counterfactual is a polynomial fitted to the density outside an excluded region around the kink, and `bunching::bunchit()` implements the estimator with its bootstrap. What bunching identifies is limited: Blomquist, Newey, Kumar, and Liang (2021) show that without a restriction on the distribution of the underlying heterogeneity the elasticity is not identified from the excess mass alone, so the polynomial counterfactual is an assumption, not a diagnostic.

10 Practice

10.1 Replications and anchors

- **Bagwe, Margitic, and Stashko (2026), polling place location.** A geographic fuzzy RD with signed distance to the precinct border as the score and polling-place distance as a continuous treatment, clustered by border segment; `rdrobust(fuzzy = )` reproduces the main table from the released files, and `rd_checks()` the balance and sensitivity figures.
- **Sejas-Portillo, Moro, and Stowasser (2025), energy labels.** Sharp RD at six label thresholds on seven million property sales, with density tests, donut holes, and an instrument for the final score; `rdmc()`, `rddensity()`, and `rd_checks()` cover the workflow, and `rd_adjust()` with property characteristics the precision.
- **Bleemer and Mehta (2022), the economics major.** Fuzzy RD on a discrete grade average with honest intervals; the student data are confidential, so the design is reproduced on simulated data with `RDHonest` and `rd_weak_iv()`.
- **Asher and Novosad (2020), rural roads.** Fuzzy RD on village population thresholds; `rdrobust(fuzzy = )`, `rd_checks()`, and `rddensity()`.
- **Landaïs (2015), unemployment benefits.** A regression kink on the benefit schedule; `rd_kink()` and the four graphs.

10.2 A reporting checklist

1. State the rule, the running variable, the cutoff, and who could manipulate the score and how precisely.
2. Show the RD plot with the estimator's own local fits, and the density and covariate plots next to it.
3. Report the robust bias-corrected estimate at the MSE-optimal bandwidth with the effective sample sizes on each side, and an honest interval for a range of smoothness bounds.
4. Report the density test, the balance table with its joint test, placebo cutoffs, the bandwidth sensitivity figure, and donut-hole estimates.
5. In fuzzy designs, report the first stage with its effective F and an Anderson-Rubin set.
6. Use covariates for precision, not identification; show the estimate with and without them.
7. State the estimand: the effect at the cutoff, for compliers if fuzzy, averaged over the window if local randomization, and what extrapolation assumption any statement beyond that relies on.

11 Appendix

11.1 Identification of the sharp RD effect

Let $Y = DY(1) + (1 - D)Y(0)$ with $D = 1\{X \geq c\}$. For $x \geq c$, $E[Y \mid X = x] = E[Y(1) \mid X = x] = \mu_1(x)$; for $x < c$, $E[Y \mid X = x] = \mu_0(x)$. If μ_1 and μ_0 are continuous at c , then $\lim_{x \downarrow c} E[Y \mid X = x] = \mu_1(c)$ and $\lim_{x \uparrow c} E[Y \mid X = x] = \mu_0(c)$, and their difference is $E[Y(1) - Y(0) \mid X = c]$. Positive density at c guarantees that both one-sided limits are limits of estimable quantities. No assumption on the joint distribution of $(Y(1), Y(0))$ beyond the two conditional means is used, and no assumption on the density's continuity is needed for the identity itself (Hahn, Todd, and van der Klaauw 2001).

11.2 Bias, variance, and the rate of the local linear estimator

Take the right side, $c = 0$, a uniform kernel on $[0, h]$, and μ_1 twice continuously differentiable. The local linear intercept is $\hat{\alpha}_+ = \sum_i w_i Y_i$ with weights w_i that reproduce linear functions exactly: $\sum_i w_i = 1$ and $\sum_i w_i X_i = 0$. Expanding $\mu_1(X_i) = \mu_1(0) + \mu_1'(0)X_i + \frac{1}{2}\mu_1''(\tilde{X}_i)X_i^2$,

$$E[\hat{\alpha}_+ \mid X] - \mu_1(0) = \frac{1}{2} \sum_i w_i \mu_1''(\tilde{X}_i) X_i^2 = O(h^2),$$

because $|X_i| \leq h$ on the support of the weights and the weights are bounded in absolute sum. The number of observations with nonzero weight is proportional to nh when the density is bounded away from zero, and

$\text{Var}(\hat{\alpha}_+ | X) = \sigma_1^2 \sum_i w_i^2 = O(1/(nh))$. The mean squared error $C_1 h^4 + C_2/(nh)$ is minimized at $h \propto n^{-1/5}$, giving $\text{MSE} \propto n^{-4/5}$, which is the minimax rate over the class of functions with bounded second derivative (Tsybakov 2009). The kernel changes the constants, not the rates.

11.3 The covariate adjustment theorem

Let $\tilde{Y} = Y - Z'\gamma_0$ with γ_0 the probability limit of the local coefficient on Z . If $E[Z | X = x]$ is twice continuously differentiable at c and Z is predetermined, then $E[\tilde{Y} | X = x] = E[Y | X = x] - E[Z | X = x]'\gamma_0$ has the same jump at c as $E[Y | X = x]$ and the same second derivatives up to a smooth term, so the local linear estimator on $\tilde{Y}$ has the bias $h^2 B$ of Section 3.3 and the variance $V_{\text{lin}} = \bar{\kappa} [\text{Var}(\tilde{Y} | X = c^+) + \text{Var}(\tilde{Y} | X = c^-)]/f_X(c)$, which is at most V (Calonico, Cattaneo, Farrell, and Titiunik 2019). Replacing $Z'\gamma_0$ by any $\eta(Z)$ with $E[\eta(Z) | X = x]$ smooth at c leaves the argument unchanged, which is the flexible adjustment of Noack, Olma, and Rothe (2024); cross-fitting removes the dependence of $\hat{\eta}$ on the observation it adjusts, so the estimated adjustment behaves as a fixed function in the local step.

11.4 Bias correction and the robust variance

Write the local linear estimator at bandwidth h as $\hat{\tau}(h)$ with $\hat{\tau}(h) - \tau \approx h^2 B + \text{noise}$, and let $\hat{B}(b)$ be the estimate of B from the difference of the local quadratic second derivatives at bandwidth b . The bias-corrected estimator is $\hat{\tau}^{bc} = \hat{\tau}(h) - h^2 \hat{B}(b)$. Its variance is $\text{Var}(\hat{\tau}(h)) + h^4 \text{Var}(\hat{B}(b)) - 2h^2 \text{Cov}(\hat{\tau}(h), \hat{B}(b))$, which the robust standard error estimates; the conventional standard error keeps only the first term. Calonico, Cattaneo, and Titiunik (2014) show that with h MSE-optimal and b chosen so that the bias of $\hat{B}$ is negligible, $(\hat{\tau}^{bc} - \tau)/\widehat{se}^{rb}$ is asymptotically standard normal, whereas $(\hat{\tau}(h) - \tau)/\widehat{se}$ converges to a normal with nonzero mean. The ratio h/b is reported by `rdrobust` as `rho`; when $b = h$ the correction and the variance adjustment take a particularly simple form and the estimator is a local quadratic fit.

11.5 Honest intervals

Let $\hat{\tau} = \sum_i w_i Y_i$ be any linear estimator with weights summing to $+1$ on the right and -1 on the left. Over the class of functions whose second derivative is bounded by M on each side, the largest possible bias is attained by the function $\pm \frac{M}{2}(x - c)^2$ and equals $\bar{b}(w) = \frac{M}{2} \sum_i |w_i| (X_i - c)^2$. With $se(w) = \sqrt{\sum_i w_i^2 \sigma_i^2}$, the interval $\hat{\tau} \pm cv_\alpha(\bar{b}/se) se$, where $cv_\alpha(t)$ is the $1 - \alpha$ quantile of $|N(t, 1)|$, has coverage at least $1 - \alpha$ for every function in the class (Armstrong and Kolesár 2018). The bandwidth that minimizes the length of this interval is the fixed-length confidence interval choice; `RDHonest` computes it, and the rule-of-thumb M is derived from a global quartic fit (Armstrong and Kolesár 2020). The same construction applies when X is discrete, since it never uses the density of X (Kolesár and Rothe 2018).

11.6 Fuzzy RD as local instrumental variables

Inside the window, let $Z = 1\{X \geq c\}$, and write the local linear reduced form $\hat{\tau}_Y$ and first stage $\hat{\tau}_D$ with influence functions ψ_Y and ψ_D (Section 3.2 gives their weights). The ratio $\hat{\tau}_{FRD} = \hat{\tau}_Y/\hat{\tau}_D$ has influence function $(\psi_Y - \tau_{FRD} \psi_D)/\tau_D$ by the delta method, so its variance is $[\text{Var}(\hat{\tau}_Y) + \tau_{FRD}^2 \text{Var}(\hat{\tau}_D) - 2\tau_{FRD} \text{Cov}(\hat{\tau}_Y, \hat{\tau}_D)]/\tau_D^2$; `rdrobust` applies the bias correction to numerator and denominator jointly. The Anderson-Rubin statistic for a candidate value τ_0 regresses $Y - \tau_0 D$ on Z and the controls with kernel weights and tests the coefficient on Z ; its distribution does not depend on τ_D , which is what makes the inverted set valid when τ_D is small (Feir, Lemieux, and Marmer 2016).

11.7 References

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